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AE 520 Compressible Flow
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Homework 2

Problem 1

$R(t)$: balloon radius; ρ_i and ρ_o: inside and outside densities. The densities are constant, and the inlet-tube volume is negligible.

$$Q = 2.0 \times 10^{-3}\ \mathrm{m^3/s}, \quad \rho_i = 1.30\ \mathrm{kg/m^3}, \quad \rho_o = 1.20\ \mathrm{kg/m^3}, \quad R_0 = 0.10\ \mathrm{m}, \quad R_c = 0.15\ \mathrm{m}.$$

(a) $\mathbf{n}$: outward unit normal; $\mathbf{u}_b$: control-surface velocity. The control surface follows the membrane and cuts the inlet tube.

$$0 = \frac{d}{dt}\int_{V_b(t)} \rho\, dV + \int_{S_b(t)} \rho(\mathbf{u}-\mathbf{u}_b)\cdot\mathbf{n}\, dA.$$

At the membrane,

$$\mathbf{u}\cdot\mathbf{n} = \mathbf{u}_b\cdot\mathbf{n} = \frac{dR}{dt} \quad \Longrightarrow \quad \rho_i(\mathbf{u}-\mathbf{u}_b)\cdot\mathbf{n} = \rho_i\left(\frac{dR}{dt}-\frac{dR}{dt}\right) = 0.$$

With Q evaluated at the inlet density ρ_i,

$$\frac{d}{dt}(\rho_i V_b) - \rho_i Q = 0, \qquad V_b = \frac{4\pi}{3}R^3,$$

$$\rho_i 4\pi R^2 \frac{dR}{dt} = \rho_i Q \qquad \Longrightarrow \qquad \boxed{\frac{dR}{dt} = \frac{Q}{4\pi R^2}}.$$

(b)

$$\left.\frac{dR}{dt}\right|_{R=0.15\,\mathrm{m}} = \frac{2.0\times10^{-3}}{4\pi(0.15)^2} = \boxed{7.07\times10^{-3}\ \mathrm{m/s}}.$$

(c)

$$\frac{dV_b}{dt} = Q \quad \Longrightarrow \quad V_b(t) - V_b(0) = Qt,$$

$$\frac{4\pi}{3}[R^3(t) - R_0^3] = Qt \quad \Longrightarrow \quad R(t) = \left(R_0^3 + \frac{3Qt}{4\pi}\right)^{1/3}.$$

Let t_c be the time at which $R(t_c) = R_c$:

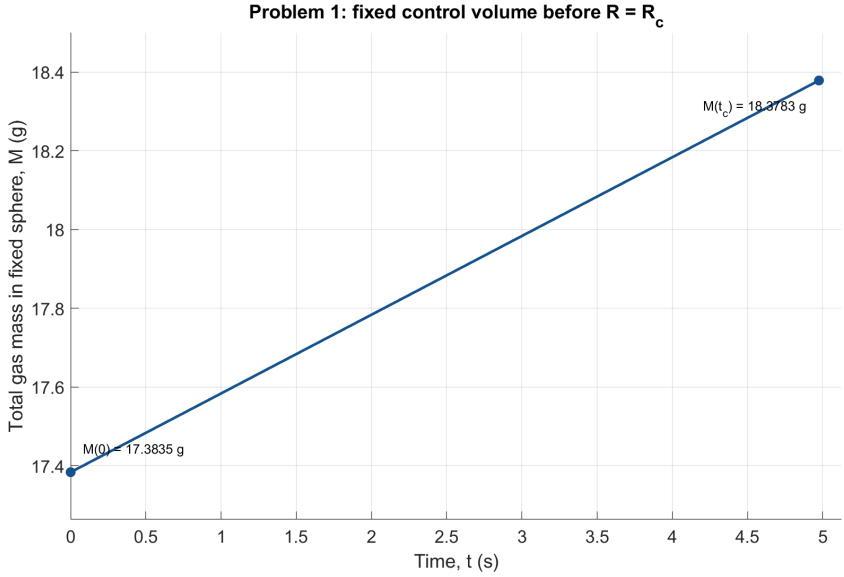
$$t_c = \frac{4\pi(R_c^3 - R_0^3)}{3Q} = \frac{4\pi[(0.15)^3 - (0.10)^3]}{3(2.0\times10^{-3})} = \boxed{4.974\ \mathrm{s}}.$$

(d) $M(t)$ is the total gas mass inside the fixed sphere, including both gases.

$$V_c = \frac{4\pi}{3}R_c^3, \qquad V_{b,0} = \frac{4\pi}{3}R_0^3,$$
$$M(t) = \rho_i V_b(t) + \rho_o[V_c - V_b(t)]$$
$$= \rho_o V_c + (\rho_i - \rho_o)(V_{b,0} + Qt).$$

$$\boxed{M(t) = 0.0173835 \text{ kg} + \left(2.00 \times 10^{-4} \text{ kg/s}\right) t, \qquad 0 \le t \le t_c.}$$

$$M(0) = 17.3835 \text{ g}, \qquad M(t_c) = \rho_i V_c = 18.3783 \text{ g}.$$



(e) For $R > R_c$, the fixed sphere contains only gas of density ρ_i. Define the outward radial component and its surface average by

$$u_r = \mathbf{u} \cdot \mathbf{n}, \qquad \overline{u}_r = \frac{1}{4\pi R_c^2}\int_{S_c} u_r \, dA.$$

$$M = \rho_i V_c = \text{constant}, \qquad \frac{dM}{dt} + \rho_i 4\pi R_c^2 \overline{u}_r - \rho_i Q = 0,$$

$$\boxed{\overline{u}_r = \frac{Q}{4\pi R_c^2} = 7.07 \times 10^{-3} \text{ m/s} \quad \text{(outward)}.}$$

The inlet is counted separately; its small opening area is neglected.

(f)

$$\text{Moving balloon volume:} \quad \frac{d}{dt}(\rho_i V_b) = \rho_i Q,$$
$$\text{Fixed sphere, } R < R_c: \quad \frac{dM}{dt} = \rho_i Q - \rho_o Q = (\rho_i - \rho_o)Q,$$
$$\text{Fixed sphere, } R > R_c: \quad \frac{dM}{dt} = \rho_i Q - \rho_i Q = 0.$$

Both satisfy mass conservation. The moving membrane has zero relative mass flux, whereas gas crosses the fixed spherical surface; the enclosed mass therefore changes differently.

Problem 2

(a) Steady, constant-density flow; all fluxes and forces are per unit span. $\mathbf{n}$ points outward.

$$0 = \oint_{CS} \rho \mathbf{u}\cdot\mathbf{n}\, dA = -\rho U H + \rho \int_0^H u(y)\, dy + \rho v_H L,$$

$$\int_0^H u(y)\, dy = \int_0^\delta \frac{Uy}{\delta}\, dy + \int_\delta^H U\, dy = \frac{U\delta}{2} + U(H-\delta) = U\left(H - \frac{\delta}{2}\right),$$

$$v_H L = UH - U\left(H - \frac{\delta}{2}\right) = \frac{U\delta}{2},$$

$$v_H = \frac{U\delta}{2L} = \frac{30(1.20\times 10^{-3})}{2(0.50)} = \boxed{+0.0360 \text{ m/s}}.$$

Since $v_H > 0$, the top flow is outward, in the positive y direction.

(b) Neglect streamwise body force and viscous tractions on the inlet, outlet, and top. Equal inlet/outlet pressures give zero net pressure force in x. $F'_{w\to f,x}$ denotes the wall force on the fluid per unit span.

$$F'_{w\to f,x} = \oint_{CS} \rho u(\mathbf{u}\cdot\mathbf{n})\, dA = -\rho U^2 H + \rho \int_0^H u^2(y)\, dy + \rho U v_H L,$$

$$\int_0^H u^2(y)\, dy = \frac{U^2}{\delta^2}\int_0^\delta y^2\, dy + U^2(H-\delta) = U^2\left(H - \frac{2\delta}{3}\right),$$

$$F'_{w\to f,x} = \rho U^2\left[-H + H - \frac{2\delta}{3} + \frac{\delta}{2}\right] = -\frac{\rho U^2 \delta}{6}$$

$$= -\frac{(1.20)(30)^2(1.20\times 10^{-3})}{6} = \boxed{-0.216 \text{ N/m}}.$$

The top carries streamwise momentum $\rho U v_H L = 0.648$ N/m. The wall force on the fluid is upstream. Its average shear magnitude is

$$\boxed{\overline{\tau}_w = -\frac{F'_{w\to f,x}}{L} = \frac{0.216}{0.50} = 0.432 \text{ Pa}}.$$

The signed average wall traction on the fluid is -0.432 Pa.

(c) Use $(\nabla\mathbf{u})_{ij} = \dfrac{\partial u_i}{\partial x_j}$: rows are velocity components and columns are coordinate directions.

$$\frac{\partial u}{\partial x} = -\frac{v_H}{\delta} = -30 \text{ s}^{-1}, \qquad \frac{\partial u}{\partial y} = \frac{U}{\delta} = 25000 \text{ s}^{-1},$$

$$\frac{\partial v}{\partial x} = 0, \qquad \frac{\partial v}{\partial y} = \frac{v_H}{\delta} = 30 \text{ s}^{-1},$$

$$\boxed{\nabla\mathbf{u} = \begin{pmatrix} -30 & 25000 & 0 \\ 0 & 30 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ s}^{-1}}.$$

$$\mathbf{D} = \frac{1}{2}\left[\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right], \qquad \mathbf{W} = \frac{1}{2}\left[\nabla\mathbf{u} - (\nabla\mathbf{u})^T\right],$$

$$\boxed{\mathbf{D} = \begin{pmatrix} -30 & 12500 & 0 \\ 12500 & 30 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ s}^{-1}}, \qquad \boxed{\mathbf{W} = \begin{pmatrix} 0 & 12500 & 0 \\ -12500 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ s}^{-1}}.$$

$$\boxed{\nabla\cdot\mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = -30 + 30 + 0 = 0}.$$

(d) $\mathbf{I}$ is the three-dimensional identity tensor.

$$\mathbf{D}' = \mathbf{D} - \frac{1}{3}(\nabla \cdot \mathbf{u})\mathbf{I} = \mathbf{D},$$
$$\boldsymbol{\tau} = 2\mu\mathbf{D}' + \zeta(\nabla \cdot \mathbf{u})\mathbf{I} = 2\mu\mathbf{D},$$
$$\boxed{\boldsymbol{\tau} = \begin{pmatrix} -0.001080 & 0.4500 & 0 \\ 0.4500 & +0.001080 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ Pa}}.$$

The nonzero stresses are τ_{xx}, τ_{yy}, and $\tau_{xy} = \tau_{yx}$. The bulk term vanishes because $\nabla \cdot \mathbf{u} = 0$.

(e)

$$\begin{aligned}\Phi = \boldsymbol{\tau} : \nabla\mathbf{u} &= \sum_{i,j} \tau_{ij} \frac{\partial u_i}{\partial x_j} \\ &= \tau_{xx}\frac{\partial u}{\partial x} + \tau_{xy}\frac{\partial u}{\partial y} + \tau_{yx}\frac{\partial v}{\partial x} + \tau_{yy}\frac{\partial v}{\partial y} \\ &= (-0.001080)(-30) + (0.4500)(25000) + (0.4500)(0) + (0.001080)(30) \\ &= \mu\left(\frac{U}{\delta}\right)^2 + 4\mu\left(\frac{v_H}{\delta}\right)^2 \simeq \boxed{1.125 \times 10^4 \text{ W/m}^3}.\end{aligned}$$

$$\boxed{\dot{s}_{\text{gen}} = \frac{\Phi}{T} = \frac{11250.0648}{300} \simeq 37.50 \text{ W/(m}^3\text{ K)}}.$$

This is viscous entropy production per unit volume.

(f)

$$\tau_{w,\text{local}} \approx \mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)_{x=L,y=0} = \frac{\mu U}{\delta} = \boxed{0.450 \text{ Pa}},$$
$$\frac{\tau_{w,\text{local}} - \overline{\tau}_w}{\overline{\tau}_w} \times 100\% = \frac{0.450 - 0.432}{0.432} \times 100\% = \boxed{4.17\%}.$$

The gradient estimate is local at the outlet; $\overline{\tau}_w$ is averaged over the plate. The velocity profiles are approximate, so exact agreement is not expected.

Problem 4

Steady, adiabatic ideal-gas flow with constant heat capacities and no shaft work. Neglect elevation change, heat conduction, and viscous work across the station faces. h, s, c_p are mass-specific; V is speed.

(a) $c_p - c_v = R$, $\gamma = c_p/c_v$.

$$c_p = \frac{\gamma R}{\gamma - 1} = \frac{1.4(287)}{0.4} = 1004.5 \text{ J/(kg K)},$$

$$V_1 = M_1\sqrt{\gamma R T_1} = 0.80\sqrt{1.4(287)(300)} = \boxed{277.75 \text{ m/s}}.$$

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} = h_0, \qquad c_p(T_0 - T) = \frac{V^2}{2}.$$

$$T_0 = T_1 + \frac{V_1^2}{2c_p} = 300\left[1 + \frac{1.4-1}{2}(0.80)^2\right] = \boxed{338.400 \text{ K}},$$

$$T_2 = T_0 - \frac{V_2^2}{2c_p} = 338.4 - \frac{100^2}{2(1004.5)} = \boxed{333.422 \text{ K}}.$$

(b) **Case A:** $s_2 - s_1 = 0$.

$$s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\frac{p_2}{p_1} = 0,$$

$$p_{2,A} = p_1\left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)} = 100\left(\frac{333.4224}{300}\right)^{3.5} = \boxed{144.730 \text{ kPa}}.$$

The stagnation state $0, i$ is defined by isentropic deceleration to rest:

$$s_{0,i} = s_i \quad\Longrightarrow\quad p_{0,i} = p_i\left(\frac{T_{0,i}}{T_i}\right)^{\gamma/(\gamma-1)}.$$

$$p_{0,1} = 100\left(\frac{338.4}{300}\right)^{3.5} = \boxed{152.434 \text{ kPa}}, \qquad \boxed{p_{0,2,A} = p_{0,1} = 152.434 \text{ kPa}}.$$

(c) **Case B:** $\dfrac{s_2 - s_1}{R} = 1$; T_0 and T_2 are unchanged.

$$\ln\frac{p_{2,B}}{p_1} = \frac{c_p}{R}\ln\frac{T_2}{T_1} - \frac{s_2 - s_1}{R} = 3.5\ln\frac{T_2}{T_1} - 1,$$

$$p_{2,B} = p_1\left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)} e^{-1} = p_{2,A}e^{-1} = \boxed{53.2430 \text{ kPa}},$$

$$p_{0,2,B} = p_{2,B}\left(\frac{T_0}{T_2}\right)^{3.5} = \boxed{56.0773 \text{ kPa}}.$$

The specified entropy increase gives $p_{2,B} < p_1$ despite $V_2 < V_1$.

(d)

$$s_{0,2} - s_{0,1} = s_2 - s_1, \qquad h_{0,2} = h_{0,1} \quad\Longrightarrow\quad T_{0,2} = T_{0,1},$$

$$s_2 - s_1 = c_p\ln\frac{T_{0,2}}{T_{0,1}} - R\ln\frac{p_{0,2}}{p_{0,1}} = -R\ln\frac{p_{0,2}}{p_{0,1}},$$

$$\boxed{\frac{p_{0,2}}{p_{0,1}} = \exp\left[-\frac{s_2 - s_1}{R}\right]}.$$

(e)

Quantity	Case A	Case B
h_0	Constant	Constant
s	Constant	Increases: $s_2 - s_1 = 287$ J/(kg K)
p_0	Constant	Decreases: $p_{0,2}/p_{0,1} = e^{-1} = 0.367879$

Problem 6

Stationary, one-dimensional adiabatic shock; no shaft work or elevation change. α is the prescribed atomic-oxygen mass fraction. Component heat capacities are constant; enthalpies and gas constants are mass-specific.

(a) Upstream density:

$$\rho_1 = \frac{p_1}{R_{\mathrm{O}_2} T_1} = \frac{10^4}{(259.84)(300)} = 0.128284 \text{ kg/m}^3.$$

With $r = \rho_2/\rho_1$,

$$\rho_1 u_1 = \rho_2 u_2 \quad \Longrightarrow \quad u_2 = \frac{u_1}{r},$$

$$p_2 = p_1 + \rho_1 u_1^2 - \rho_2 u_2^2 = p_1 + \rho_1 u_1^2 \left(1 - \frac{1}{r}\right)$$

$$\simeq \boxed{\rho_1 u_1^2 \left(1 - \frac{1}{r}\right)} \qquad (p_1 \ll \rho_1 u_1^2).$$

$$p_2 = \rho_2 R_2 T_2 \quad \Longrightarrow \quad T_2 = \frac{p_2}{r \rho_1 R_2} \simeq \boxed{\frac{u_1^2 (r-1)}{r^2 R_2}}.$$

(b)

$$R_2 = (1-\alpha) R_{\mathrm{O}_2} + \alpha (2 R_{\mathrm{O}_2}) = R_{\mathrm{O}_2}(1+\alpha),$$

$$c_{p,2} = (1-\alpha)\frac{7}{2} R_{\mathrm{O}_2} + \alpha \frac{5}{2}(2 R_{\mathrm{O}_2}) = R_{\mathrm{O}_2}\left(\frac{7}{2} + \frac{3}{2}\alpha\right),$$

$$C \equiv \frac{c_{p,2}}{R_2} = \frac{3.5 + 1.5\alpha}{1+\alpha}, \qquad \beta \equiv \frac{\alpha Q_d}{u_1^2}.$$

Energy conservation, with $h_1 \ll u_1^2/2$:

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \quad \Longrightarrow \quad \frac{u_1^2}{2} \simeq c_{p,2} T_2 + \alpha Q_d + \frac{u_1^2}{2 r^2}.$$

Substituting T_2 from (a) and dividing by u_1^2,

$$\frac{1}{2} = C \frac{r-1}{r^2} + \beta + \frac{1}{2r^2} \quad \Longrightarrow \quad r^2 = 2C(r-1) + 2\beta r^2 + 1,$$

$$\boxed{(1-2\beta) r^2 - 2Cr + (2C-1) = 0.}$$

Select the compression root $r > 1$:

$$\boxed{r = \frac{C + \sqrt{C^2 - (1-2\beta)(2C-1)}}{1-2\beta}.}$$

(c) Using $u_1 = 5000$ m/s and $Q_d = 15.6 \times 10^6$ J/kg,

$$\beta = 0.624\alpha, \qquad R_2 = 259.84(1+\alpha) \text{ J/(kg K)}.$$

α	β	r	T_2 (K)	p_2 (MPa)
0.00	0.0000	6.0000	13362.9	2.6726
0.10	0.0624	6.6081	11233.2	2.7218
0.20	0.1248	7.4912	9274.2	2.7790
0.30	0.1872	8.7906	7461.5	2.8423

(d) As α increases, more energy is stored in αQ_d, leaving less for $c_{p,2} T_2$, so T_2 decreases. The table also shows greater compression (r increases) and therefore lower downstream speed, $u_2 = u_1/r$.